

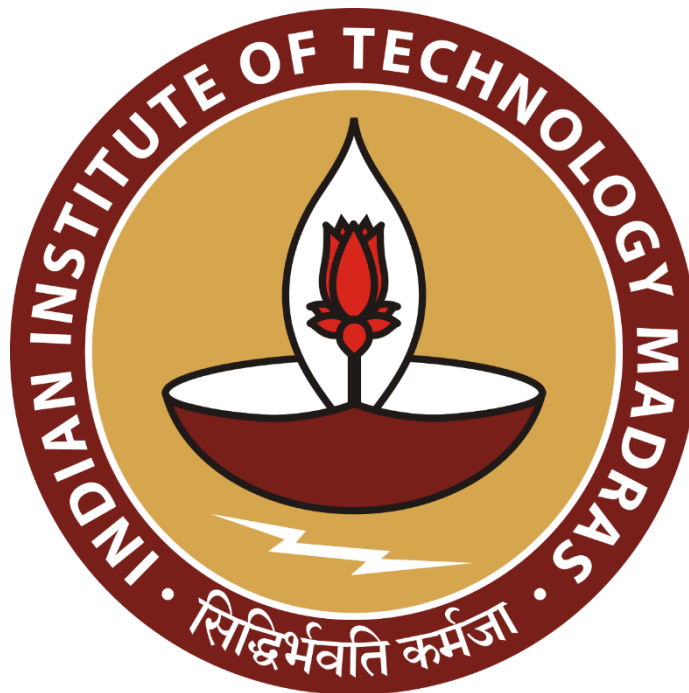
Optimal Inventory Management and Sales Forecasting for Janananma Pharmacy

A Final Report for the BDM Capstone Project

Submitted by

Name: **Nihila Pallath**

Roll number: **22F2001240**



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

Contents

1	Executive Summary and Title	2
2	Detailed Explanation of Analysis Process/Method	3
3	Results and Findings	4
4	Detailed Explanation of Analysis Process/Method	7
5	Interpretation of Results and Recommendation	9

1. Executive Summary and Title

Title: Optimal Inventory Management and Sales Forecasting for Janananma Pharmacy

Janananma is a B2C leading healthcare provider committed to deliver quality pharmaceutical products and services to customers. It established in 2018, the pharmacy located in Mannarkkad, Palakkad district Kerala. The pharmacy offers a comprehensive range of prescription and over-the-counter medications, wellness products, and surgical products. Janananma facing key operational challenges that impact its efficiency and profitability. These challenges include inventory management issues, where maintaining optimal stock levels has been difficult, sometimes they face stockout issues and sometimes more expense for unnecessary stocks; expiry management, where products nearing their expiry dates are not being utilized or disposed of efficiently and keep more stocks than required also leads to expire the products; and brand diversity, which complicates procurement and sales strategies. Addressing these problems is essential to reducing financial losses, enhancing customer satisfaction, and improving operational efficiency.

To address these concerns, the data gathered from Remedy Plus software is used by the pharmacy to collect the sales, purchase and inventory details. This project addresses the critical challenge of stock management and sales forecasting for Janananma Medicals, a pharmacy aiming to optimize inventory levels while minimizing the risk of medicine stockouts and overstocking. Using sales, purchase, inventory, and expiry data, it's implemented a data-driven approach to analyze current inventory dynamics and forecast future stock requirements. The analysis based on 19,021 sales records and 3,447 purchase records 01/10/2023 to 31/03/2024 and collected from Remedy Plus software which is used by this pharmacy to store the data. Python is the main tool used to do the analysis with the help of Pandas, Matplotlib.

At the earlier it was shortlisted the data based on the most sold products to do the stock management and sales forecast. And the forecast is done based on the ARIMA method. As the understanding of considering only the sold quantity to shortlist the data is not a good method, it begun by identifying common products across sales and purchase data, based on the high profitable medicines per unit as well as high profitable medicines included its quantity of sold by using the purchase and sales data, ensuring that analysis was focused on actively traded medicines. For each unique combination of Product and Batch Number, calculated average unit cost, sales and purchase quantities, and profit per unit. This enabled to identify top-performing and loss-making products based on unit profitability.

To ensure that essential medicines remain in stock, developed a Reorder Point (ROP) model based on average daily usage and lead times. This allowed to determine which medicines were at risk of

stockout and needed urgent reordering. Also computed the Economic Order Quantity (EOQ) to optimize ordering frequency and reduce holding costs.

Furthermore, leveraging time-series forecasting techniques—including Moving Average, ARIMA, and Exponential Smoothing forecasted future demand for each product. Comparative analysis on selected products, such as *ACENOMAC 2 MG*, illustrated how each method performs over upcoming months.

Finally, the integrated analysis calculated total profit and current stock levels, merged with expiry dates and supplier data to ensure insights were practical and actionable. The outcome is a comprehensive reporting and forecasting tool that empowers Janananma Medicals to make data-backed decisions, improve profit margins, and maintain optimal stock levels for patient care.

2. Detailed Explanation of Analysis Process/Method

Data Preprocessing

To ensure accurate and meaningful analysis, the data collected from the duration of 6 months underwent thorough preprocessing. First, the raw sales, purchase, and inventory datasets were cleaned by handling missing values and standardizing formats. The data was in unstructured form, by using the macro tool in the excel transferred data into a structured format.

JANANANMA MEDICALS & SURGICALS						
HSPTL JUNC , NEAR KPM HOTEL, MANNARKKAD, Whatsapp : 8129395919						
Bill Statement						
From : 01/10/2023 To : 31/03/2024					page no	1
bill n	17895		mode	CASH		
			date	01/10/2023		
patien	VARGHESE		dr. :	SURESH		
product	batchno	expiry		dis%		tax%
GLUVILDA M500	GLD23013MH	03/2025		10.00		12.00
CLOPIVAS AP-150	EC0313	03/2025		10.00		12.00
VOGLET M0.3 TAB	V2275601	10/2024		10.00		12.00
PROLOMET XL-25 TAB	GTE1096A	04/2025		10.00		12.00
MONIT GTN 2.6 TAB	N2301333	04/2025		10.00		12.00
PANTOP.D	SPD230449	03/2025		8.00		12.00
BIOPRIDE 2 TAB	ajl23008	01/2025		10.00		12.00
DIGEPLEX-T	DTSC004	03/2024		7.00		12.00
ROSUMAC-5mg tab	KRK2301A	02/2026		10.00		12.00
D-PROTIEN 200GM VANIL	DPVA220801	07/2024		3.50		12.00
MURIVENNA	4665	07/2024		0.00		5.00

Table 1: Row Sales Data

Patient Name	Doctor Name	Date	Product	Batch No.	Expiry Date	Qty	Rate	Discount	Amount	Tax	Total	Year Month
AYSHA	RANJITH	01-10-2023	ACILOC 150 TAB 30,S	LD22304	31-10-2024	4	1.3647	4	5.24	12	5.895056	2023-10
ASHNA	189	01-10-2023	AEROCORT ROTOCAP	SA30955	31-10-2024	1	114.35	10	102.92	12	116.642	2023-10
JANU	AJAY K GOPAL	01-10-2023	ANGICAM BETA TAB 15,	agb2323	30-04-2025	2	2.5333	2	4.97	12	5.577992	2023-10
YOUSUF KARU	ANISH AHAMM	01-10-2023	APIDRA VIAL	2F004A	31-10-2024	1	1534	14	1319.24	5	1395.94	2023-10
RENU	ABDUL RIYAD	01-10-2023	BABY DIAPER RASH CRE	852200577	31-08-2025	2	80	0	160	12	179.2	2023-10
RENU	ABDUL RIYAD	01-10-2023	BABY DIAPER RASH CRE	282	31-03-2026	1	80	0	80	12	89.6	2023-10
patient3606	doctor3606	01-10-2023	BAND-AID	C647	31-03-2024	7	2	0	14	12	15.68	2023-10
SHAMNA	NIBIT PURKAY	01-10-2023	BETADINE-OINT	PD0813	31-03-2025	1	145.14	0	145.14	12	162.5568	2023-10
MUHAMMEDA	RANJITH	01-10-2023	BILASURE 20	AXA0025	31-01-2025	7	13.9	3	94.38	12	106.056	2023-10
VARGHESE	SURESH	01-10-2023	BIOPRIDE 2 TAB	ajl23008	31-01-2025	30	5.42	10	146.34	12	165.852	2023-10
VASUDEVAN79	RAISON P CHA	01-10-2023	BIOTOR CV 10	LE223052	30-04-2025	5	10.45	3	50.68	12	56.95	2023-10
SARATH	SURESH	01-10-2023	BRIV+50	BRITB2023A	30-09-2024	30	15.4	7	429.66	12	485.1	2023-10

Table 2: Cleaned Sales Data

The cleaned sales data contains 50890 rows and 12 columns. From this unnecessary columns, such as patient name, doctor name, and Tax, Batch No., Discount, were removed. And ‘YearMonth’ was added to group sales and purchases by monthly trends, making it easier to analyze seasonality and demand fluctuations. The columns Tax, Discount are irrelevant for the further processing of analysis because, column total already includes all these information by,

$$\text{Total} = \text{Amount} + (\text{Tax}/100) * (\text{Quantity} * \text{Rate})$$

JANANANMA MEDICALS & SURGICALS										
HSPTL JUNC , NEAR KPM HOTEL, MANNARKKAD, Whatsapp : 8129395919										
Purchase Statement - Productwise										
From : 01/10/2023 To : 31/03/2024										Page No 1
SI No	A/cDate	Bill No	BillDat	Supplier	BatchNo	Expiry	Qty	Amount	TaxAmt	Total
4-QUIN LOT EYE DROP										
1740	24/11/2023	21590	23/11/2023	ATTUPURAM DRUGS	23156	08/2025	1	138.57	16.63	155.20
							1	138.57	16.63	155.20
4-QUIN-DROPS										
1740	24/11/2023	21590	23/11/2023	ATTUPURAM DRUGS	A-3048	07/2025	2	216.17	25.94	242.11
							2	216.17	25.94	242.11
A TO Z SYP										
1918	21/12/2023	86870	19/12/2023	KIMS MEDICAL	23660073	02/2025	1	124.95	22.49	147.44
2199	03/02/2024	49813	02/02/2024	KANMONEY PHARMA	23660081	02/2025	1	126.23	22.72	148.95
							2	251.18	45.21	296.39

Table 3: Row Purchase Data

Product	A/cDate	Bill No	BillDat	Supplier	BatchNo	Expiry	Qty	Amount	TaxAmt	Total
4-QUIN-DROPS	16-05-2024	ADF4114	16-05-2024	ATTUPURAM D	DO3H003	31-07-2025	2	216.17	25.94	242.11
A TO Z NS DROPS	06-05-2024	7352	04-05-2024	SUNANDA ASSC	23079AZPF	31-01-2025	2	95.6	17.21	112.81
A TO Z NS+	15-05-2024	5443	15-05-2024	NEDSONS AGEN	24490008	30-06-2025	15	100.88	18.16	119.04
A TO Z NS+	15-05-2024	7278	15-05-2024	KANMONEY PH	23491286	31-05-2025	45	308.94	55.61	364.55
AB FLO CAP 15'S	09-04-2024	300442	08-04-2024	REVATHEEJAYA	006K23RA	31-10-2025	45	507.28	60.87	568.15
ABAXIS 5 TABLET	28-04-2024	8112	25-04-2024	GETWELL MC I	N2400081	31-12-2026	30	447.87	53.74	501.61
ABEVIA 200 SR10	07-04-2024	1213	06-04-2024	V.J PHARMA	K3GKW008	31-10-2025	20	274.37	32.92	307.29
AB-FLO SR TAB	17-04-2024	10282	15-04-2024	GETWELL MC I	11L230A	30-11-2025	20	385.72	46.29	432.01
AB-VAS-F	01-04-2024	109	01-04-2024	MONEY PHARM	ct41759	31-12-2025	30	163.16	19.58	182.74
AB-VAS-F	08-05-2024	10375	07-05-2024	MONEY PHARM	ct41759	31-12-2025	30	163.16	19.58	182.74
ABVIDA M 50/500 15S	06-05-2024	7352	04-05-2024	SUNANDA ASSC	AMS23015	30-06-2025	15	171.09	20.53	191.62
ACCU-CHEK ACTIVE 50,S	12-05-2024	13355	11-05-2024	FATIMA PHARN	29511733	30-06-2025	2	1693.26	203.19	1896.45
ACENEXT-P TAB	21-04-2024	645	19-04-2024	NATIONAL PHA	sp230101	30-09-2025	60	58.8	7.06	65.86

Table 4: Cleaned Purchase Data

The cleaned purchase data contains 3446 rows and 11 columns. From this unnecessary column, such as Bll No, BatchNo, TaxAmt, Amount were removed. And ‘YearMonth’ was added to group sales and purchases by monthly trends, making it easier to analyze seasonality and demand fluctuations. And recreated the columns A/cDate and BillDat as Received Date and Order Date.

The cleaned and filtered purchase data (Table 5) contains 3446 rows and 10 columns.

Product	A/cDate	Bill No	BillDat	Supplier	BatchNo	Expiry	Qty	Amount	TaxAmt	Total
4-QUIN-DROPS	16-05-2024	ADF4114	16-05-2024	ATTUPURAM DRUGS	DO3H003	31-07-2025	2	216.17	25.94	242.11
A TO Z NS DROP	06-05-2024	7352	04-05-2024	SUNANDA ASSOCIAT	23079AZPF	31-01-2025	2	95.6	17.21	112.81
A TO Z NS+	15-05-2024	5443	15-05-2024	NEDSONS AGENCIES	24490008	30-06-2025	15	100.88	18.16	119.04
A TO Z NS+	15-05-2024	7278	15-05-2024	KANMONEY PHARMA	23491286	31-05-2025	45	308.94	55.61	364.55
AB FLO CAP 15'	09-04-2024	300442	08-04-2024	REVATHEEJAYA MED	006K23RA	31-10-2025	45	507.28	60.87	568.15
ABAXIS 5 TABL	28-04-2024	8112	25-04-2024	GETWELL MC DIVISI	N2400081	31-12-2026	30	447.87	53.74	501.61
ABEVIA 200 SR	07-04-2024	1213	06-04-2024	V.J PHARMA	K3GKW008	31-10-2025	20	274.37	32.92	307.29
AB-FLO SR TAB	17-04-2024	10282	15-04-2024	GETWELL MC DIVISI	11L230A	30-11-2025	20	385.72	46.29	432.01
AB-VAS-F	01-04-2024	109	01-04-2024	MONEY PHARMA	ct41759	31-12-2025	30	163.16	19.58	182.74
AB-VAS-F	08-05-2024	10375	07-05-2024	MONEY PHARMA	ct41759	31-12-2025	30	163.16	19.58	182.74
ABVIDA M 50/5	06-05-2024	7352	04-05-2024	SUNANDA ASSOCIAT	AMS23015	30-06-2025	15	171.09	20.53	191.62
ACCU-CHEK AC	12-05-2024	13355	11-05-2024	FATIMA PHARMA	29511733	30-06-2025	2	1693.26	203.19	1896.45
ACENEXT-P TAB	21-04-2024	645	19-04-2024	NATIONAL PHARMA	sp230101	30-09-2025	60	58.8	7.06	65.86
ACENOMAC 2 M	02-05-2024	20857	30-04-2024	GETWELL DH DIVISI	FNV02305A	31-07-2025	70	597.52	71.7	669.22
ACENOMAC 2 M	15-05-2024	31945	14-05-2024	GETWELL DH DIVISI	FNV02307A	31-10-2025	10	85.36	10.24	95.6

Table 5: Filtered Purchase Data

Analysis based on sample data set

To better understand the financial performance of the business, created this revenue graph (Figure 1) to analyse the monthly revenue trends over time. This visualization helps identify patterns in sales, seasonal fluctuations, and potential areas for improvement.

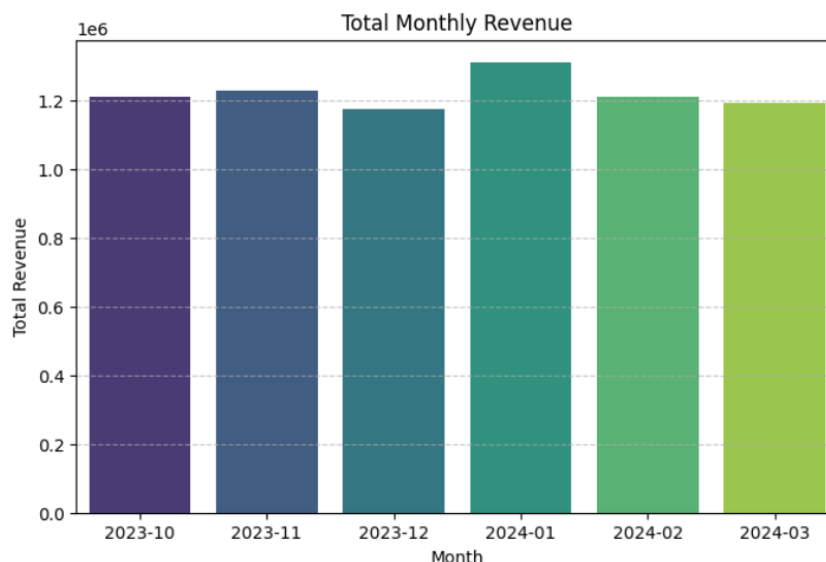


Figure 1: Monthly Revenue

From the graph, observe that revenue remains relatively stable across months, with a small peak in January 2024. This could indicate higher sales due to increased demand, special promotions, or price adjustments. In contrast, December 2023 shows a slight dip, possibly due to lower demand, end-of-year discounts, or stock shortages. Understanding these fluctuations allows us to optimize future sales strategies, ensuring we capitalize on high-revenue periods while addressing potential gaps during slower months.

A threshold of 4,000 sales was set as the benchmark, with products falling below this threshold excluded from further analysis. This filtering process was implemented to eliminate less impactful

data points and concentrate on high-performing products that contribute significantly to the pharmacy's revenue and inventory management.

The monthly sales trend analysis (Figure 2) for each medicine reveals key insights into demand patterns. The DERIPHYLLIN RETARD 150 TAB 30,S consistently records the highest sales, indicating its strong demand and essential usage. Medicines like CIFRAN 250 TAB and NEUROBION FORTE 30,S maintain relatively stable sales over time, suggesting a consistent customer base. However, certain medicines, such as PARACETAMOL 650, exhibit fluctuations, which may be influenced by seasonal demand or supply chain issues. A slight decline in sales for some medicines in early 2024 suggests potential factors such as decreased demand, stock shortages, or shifting consumer preferences. Meanwhile, moderately selling medicines like OMEE 20 CAP 20,S and GLISIMET M2 (14) show steady demand, indicating their continued relevance. Identifying these trends is crucial for inventory management, ensuring that high-demand medicines are adequately stocked while investigating the decline in sales for certain products.

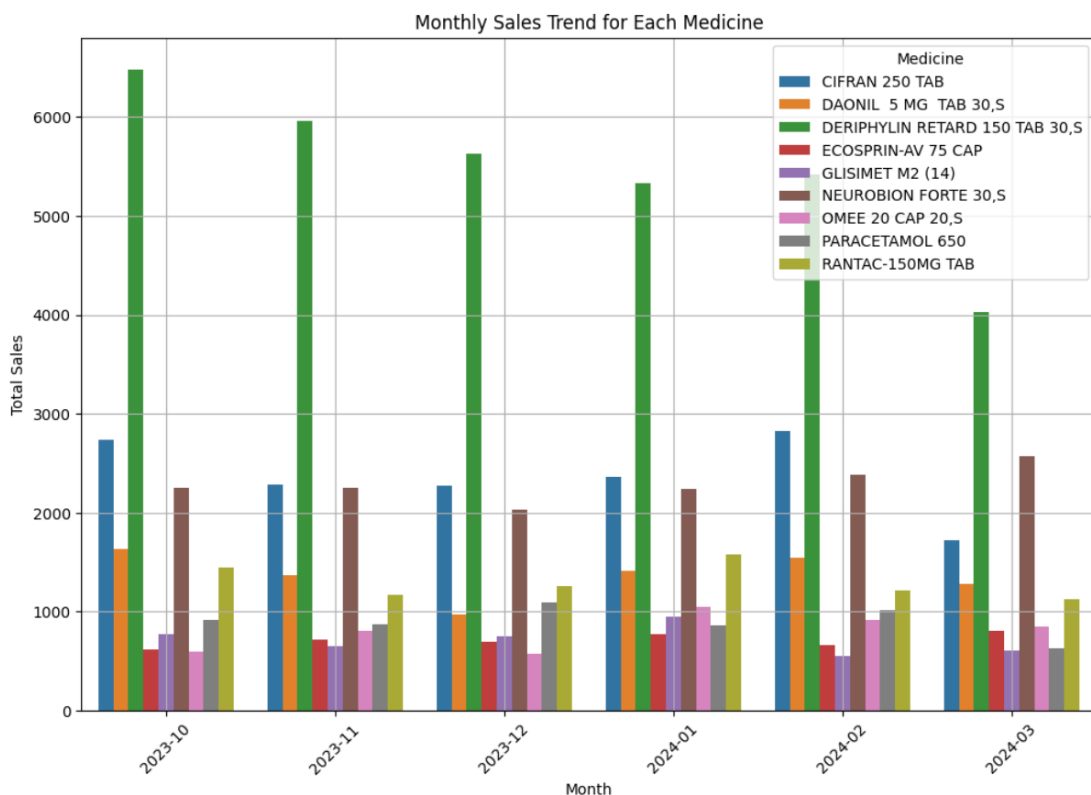


Figure 2: Monthly Sales Trend for each Products

Detailed analysis based on complete data for inventory management and sales forecast

The initial analysis has been carried out by considering the top medicines with maximum sales movement. The analysis in the upcoming sessions considers the complete data sets of medicines in order of profit margin.

1. Data Cleaning and Standardization

Removed the unwanted columns from the purchase and sales data. The columns removed are

'Amount', 'Tax', 'Rate', 'Patient Name', 'Doctor Name', 'Discount' and the columns created are 'UnitCost' which is the Total / Qty in sales data.

	Date	Product	BatchNo	Expiry Date	Qty	Total	YearMonth	UnitCost
0	2024-01-27	4-QUIN-DROPS	A-3048	31-07-2025	1	160.68	2024-01	160.680
1	2023-10-27	A TO Z SYP	22490786	31-03-2024	1	191.84	2023-10	191.840
2	2024-02-01	A TO Z SYP	23660073	28-02-2025	1	207.10	2024-02	207.100
3	2024-03-20	A TO Z SYP	23660081	28-02-2025	1	207.10	2024-03	207.100
4	2024-02-03	A TO Z NS+	23490075	30-06-2024	15	146.64	2024-02	9.776

Table 6: Processed Sales Data

In purchase data the removed columns are 'Bill No', 'Amount', 'TaxAmt' and a new column 'UnitCost' is added based on the above formula.

	Product	A/cDate	BillDat	Supplier	BatchNo	Expiry	Qty	Total	UnitCost
0	4-QUIN-DROPS	16-05-2024	16-05-2024	ATTUPURAM DRUGS	DO3H003	31-07-2025	2	242.11	121.055000
1	A TO Z NS DROPS	06-05-2024	04-05-2024	SUNANDA ASSOCIATES	23079AZPF	31-01-2025	2	112.81	56.405000
2	A TO Z NS+	15-05-2024	15-05-2024	NEDSONS AGENCIES	24490008	30-06-2025	15	119.04	7.936000
3	A TO Z NS+	15-05-2024	15-05-2024	KANMONEY PHARMA	23491286	31-05-2025	45	364.55	8.101111
4	AB FLO CAP 15'S	09-04-2024	08-04-2024	REVATHEEJAYA MEDICALS	006K23RA	31-10-2025	45	568.15	12.625556

Table 7: Processed Purchase Data

Column name 'BatchNo' is standardized. And Product names were stripped of whitespace and converted to uppercase to ensure uniformity. Date columns were converted to datetime formats for temporal analysis.

2. Identification of Common Products

For further calculations and analysis consider only the products which are in both sales and purchase. Because to find the profit need both purchase data of a product like purchase cost and expiry of that product and sales data of products like sales cost and sold date etc. So, A list of common products was generated using an inner merge between the sales data and purchase data on Product and 'BatchNo'. All subsequent analysis was limited to these common products to maintain relevance.

3. Unit Cost and Profitability Analysis

Grouped data by Product and BatchNo to calculate average UnitCost from both sales and purchase records. Because each product is purchased from one or more sellers, the purchase cost is also different for the same products that's why considering BatchNo is important to consider for finding the unique products. Profit per unit was calculated as:

$$\text{Profit} = \text{avg_sales_unit_cost} - \text{avg_purchase_unit_cost}$$

Products were categorized into:

Top Profitable Medicines: profit $\geq$ 100% of purchase cost

Least Profitable Medicines: profit $\leq$ 50% of purchase cost)

4. Stock Quantity Analysis

Total quantities purchased and sold were aggregated per product and batch. Current stock was computed as:

$$\text{current_stock} = \text{total_purchased_qty} - \text{total_sold_qty}$$

By using the inventory data which contains the current inventory (01/04/2024) as the final stock reference and sales quantity and purchase quantity, create an initial inventory list.

5. Inventory Management

By applying Economic Order Quantity (EOQ) and Reorder Point (ROP), Its aimed to create a more efficient inventory management system for the selected products. ROP ensures that orders are placed at the right time by considering the lead time and demand, preventing stockouts and ensuring continuous availability. These calculations help streamline stock replenishment, reduce holding costs, and enhance overall supply chain efficiency for the pharmacy. ROP can be calculated by using the formula,

$$\text{Reorder Point} = \text{Lead Time Demand} + \text{Safety Stock}$$

Where,

Lead Time Demand is the expected demand during the lead time. Can be calculated by,

$$\text{Lead Time Demand} = \text{Average Sales (or Demand)} \times \text{Average Lead Time.}$$

Safety Stock is the extra inventory kept preventing stockouts due to demand fluctuations and supplier lead time variability. This can be calculated by using the formula,

$$\text{Safety Stock} = (\text{Maximum Daily Sales} \times \text{Maximum Lead Time}) - (\text{Average Daily Sales} \times \text{Average Lead Time})$$

The Reorder Point ensures reorder stock in time to avoid stock outs.

In the collected data the average lead time is around 1.5 always. So took 1 as the lead time for all products on average. And the simplified ROP formula is

$$\text{ROP} = \text{Average Daily Usage} \times \text{Lead Time}$$

Usage was estimated by historical sales data grouped by month.

Products with current stock $<$ ROP were flagged as needing reorder.

EOQ helps in determining the ideal order quantity that balances ordering frequency and storage costs, ensuring that medicines are neither overstocked nor understocked. The EOQ can be determined by using the formula,

$$EOQ = \sqrt{\frac{2DS}{H}}$$

Where,

D – Annual Demand, S – Ordering cost per order, H – Holding cost per unit

Assumed constants for ordering and holding cost were used for standardization. Ordering cost as 100 per order. And holding cost as 10% of unit cost of product. EOQ helped determine the most economical quantity to restock per order.

6. Expiry Risk Monitoring

Expiry dates were used to identify medicines nearing their shelf life.

Products with high stock levels but near expiry were flagged as high-risk.

7. Sales Forecasting

Forecasts were generated using three time-series techniques:

- Moving Average
- ARIMA (Auto-Regressive Integrated Moving Average)
- Exponential Smoothing
- **Moving Average:** The Moving Average method calculates the average of sales over a fixed number of past periods and uses that as the forecast for the next period.

Considered the forecast based on past 3 month moving average. So,

$$\text{Forecast for April} = (\text{sales in January} + \text{sales in February} + \text{sales in March}) / 3$$

The moving average method is used because the data is relatively stable and has no clear trend or seasonality. It smooths out short-term fluctuations and highlights longer-term trends.

- **ARIMA (Auto-Regressive Integrated Moving Average):** ARIMA is a powerful statistical model that uses a combination of past values (Auto-Regressive), past forecast errors (Moving Average), and differencing to make time series data stationary. AR part uses the dependency between an observation and a number of lagged observations. I (Integrated) involves differencing the observations to make the time series stationary. MA part models the error of the prediction as a linear combination of error terms from the past. The ARIMA is used in this data because the data is time series depends but no strong seasonality.
- **Exponential Smoothing:** Exponential Smoothing assigns exponentially decreasing weights to older data. Recent observations have more influence on the forecast. It uses a smoothing factor (alpha, between 0 and 1). The forecast is,

$$\text{Forecast} = (\text{alpha} \times \text{current observation}) + ((1 - \text{alpha}) \times \text{previous forecast})$$

Where the data used in this forecast is automatically estimates the best value of alpha using maximum likelihood estimation, because the alpha is not manually given. As the above cases, the data has no clear trend or seasonality, using these like simple forecasting methods are better than the machine learning techniques like predictions. By using this exponential smoothening do the forecast for next 6 months of sales quantity for each products.

Monthly sales were aggregated, and forecasts were made for future demand.

Comparison of methods was visualized for a sample product (*ACENOMAC 2 MG*).

3. Results and Findings

Common Products Analysis

Identified the medicines that are present in both the sales and purchase datasets by matching based on Product and BatchNo. The BatchNo represents the different purchase cost. This step narrowed our scope to the most relevant and actively transacted products for focused analysis. Selecting this product based on product and BatchNo which is available in both sales and purchase data, there are 435 unique medicines are there in the common_product table. These 435 products are considered for further processes.

Profitability Overview

Profitability based on unit cost

Calculated profit per unit for each product as the difference between average unit sale price and purchase price. Products were grouped by Product and BatchNo to ensure batch-level granularity.

Figure 6 represents the profit table which contains each product's profit per unit.

	Product	BatchNo	avg_sales_unit_cost	avg_purchase_unit_cost	profit
0	ACENOMAC 2 MG	FNV02305A	12.18900	9.560286	2.628714
1	ACIVIR 400DT	4SN0095	15.60376	11.994500	3.609260
2	ACIVIR 400DT	4SN0100	15.60376	11.627333	3.976427
3	ADULT WIPES 10S	I0201/24	483.12000	379.996667	103.123333
4	ADULT WIPES 10S	I020224	483.12000	379.995000	103.125000

Table 9: Profit Table

From this will get that the most profitable product based on the profit per unit is,

Product : UNDERPAD PREMIUM

BatchNo : 11

avg_sales_unit_cost : 1364.0

avg_purchase_unit_cost : 170.002667

profit : 1193.997333

UNDERPAD PREMIUM | Batch: 11 stands out massively with a profit per unit nearing ₹1200, which is several times more than all others. This suggests either it's an outlier or It's a high-cost, premium product (possibly imported). So going deep into this product data,

[15] sales_data[(sales_data['Product'] == 'UNDERPAD PREMIUM') & (sales_data['BatchNo'] == '11')]									
	Date	Product	BatchNo	Expiry Date	Qty	Total	YearMonth	UnitCost	
47696	2024-03-22	UNDERPAD PREMIUM	11	31-12-2028	5	6820.0	2024-03	1364.0	

[20] purchase_data[(purchase_data['Product'] == 'UNDERPAD PREMIUM') & (purchase_data['BatchNo'] == '011')]											
	Product	A/cDate	Bill No	BillDat	Supplier	BatchNo	Expiry	Qty	Amount	TaxAmt	Total
3221	UNDERPAD PREMIUM	45386.0	0004/23-24	45383.0	MED-CARE SURGICALS	011	47118.0	5.0	720.35	129.66	850.01
3222	UNDERPAD PREMIUM	45411.0	24-25	45408.0	MED-CARE SURGICALS	011	47118.0	3.0	432.21	77.80	510.01

Figure 3: Sales and Purchase data of UNDERPAD PREMIUM

Figure 3 shows each unit was sold at ₹1364. In purchase data total 8 units were purchased for ₹1360.02, resulting in:

$$\text{Avg. Unit Purchase Cost} = ₹1360.02 / 8 = ₹170.0025$$

$$\text{Profit per Unit} = ₹1364 - ₹170 = ₹1194 \text{ approx}$$

So the profit calculation is correct. Based on the discussion with the shop owner, it is clear that this is a data entry mistake. So removed this product from the further analysis

The 2 bar charts (figure 4 and 5) represents the Medicines vs Profit per unit. x- axis represents the medicines names with its batch number. And y-axis represents the profit per unit from sales cost per unit and purchase cost per unit.

Top 20 Profitable Medicines:

Figure 4 showed the highest earning products per unit sold. (Except the outlier product - UNDERPAD PREMIUM)

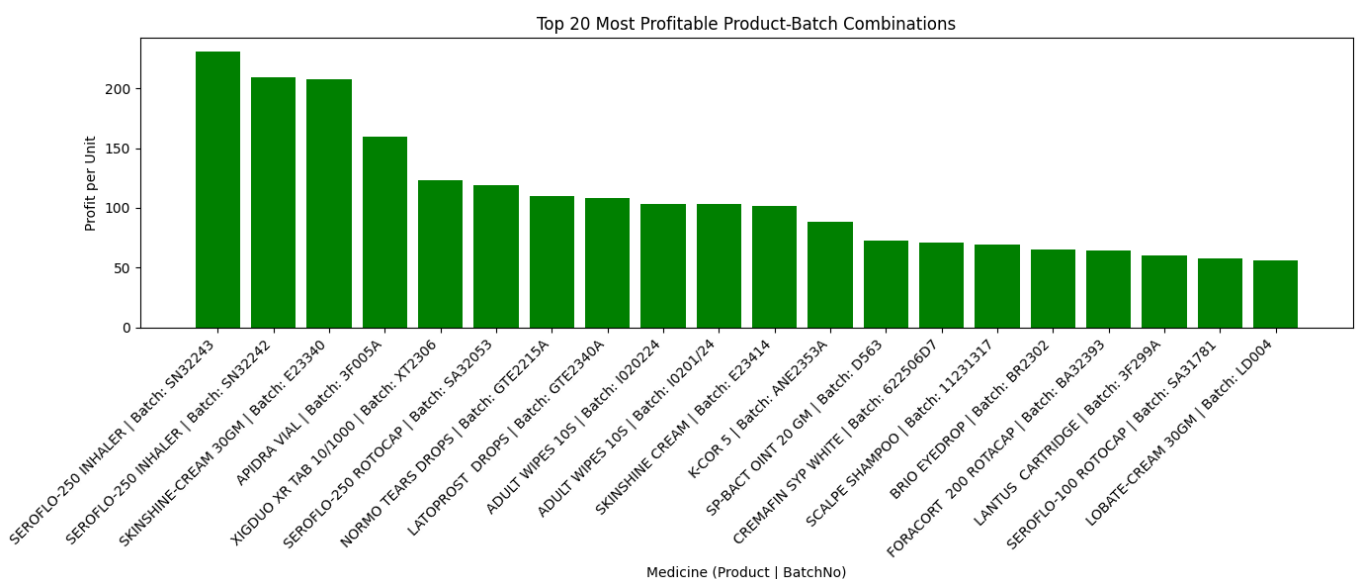


Figure 4: 20 Most Profitable Products

The products, including common items like *SEROFLO* and *XIGDUO*, show healthy but more moderate profit margins, making them strong candidates for prioritization in inventory and procurement strategies.

Bottom 20 (Low Margin Medicines):

A corresponding chart highlighted the products that are sold at or near a loss

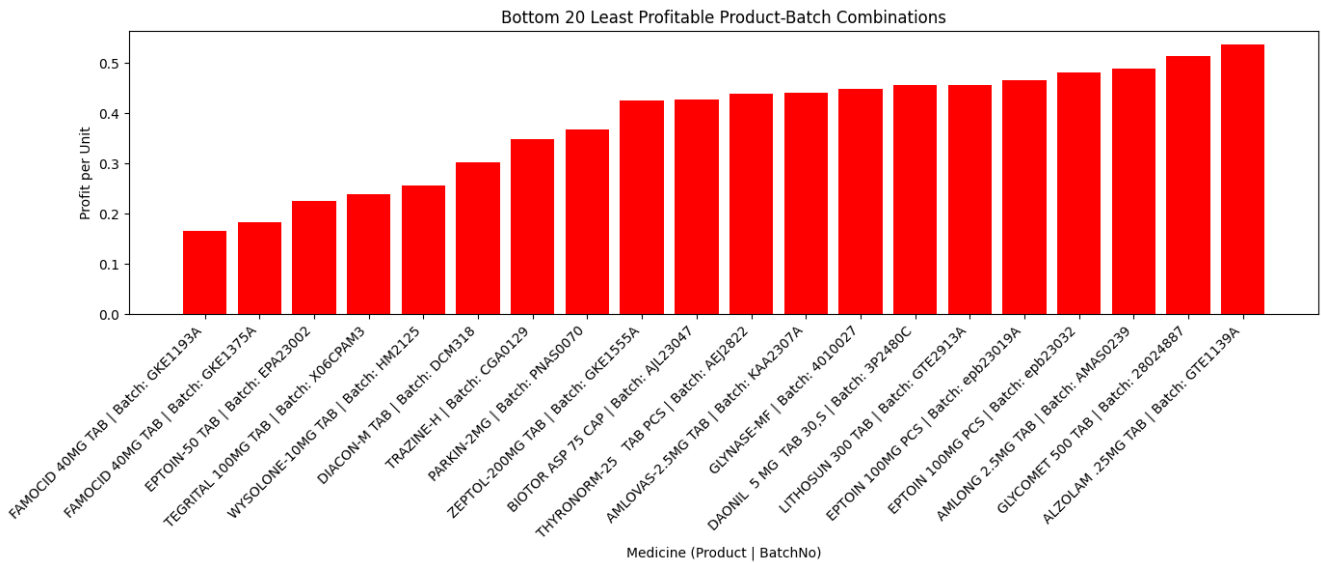


Figure 5: 20 Least Profitable Products

From the figure 5, the following insights can be made:

- The profit per unit for these medicines' ranges from ₹0.15 to ₹0.55, which is extremely low.
- Some examples from the chart:
 - FAMOCID 40MG TAB | GKE139-A — lowest profit (₹0.15/unit approx.)
 - ALZOLAM 25MG TAB | GTEL139-A — highest among the bottom (₹0.55/unit approx.)
- Products like FAMOCID, EPTOIN, WYSOLONE, and TEGRETAL appear multiple times, indicating consistently low margins across their batches.

The reason for low profit margins can be high purchase cost. So can be verify it by visualizing the sales and purchase unit cost for least portable products.

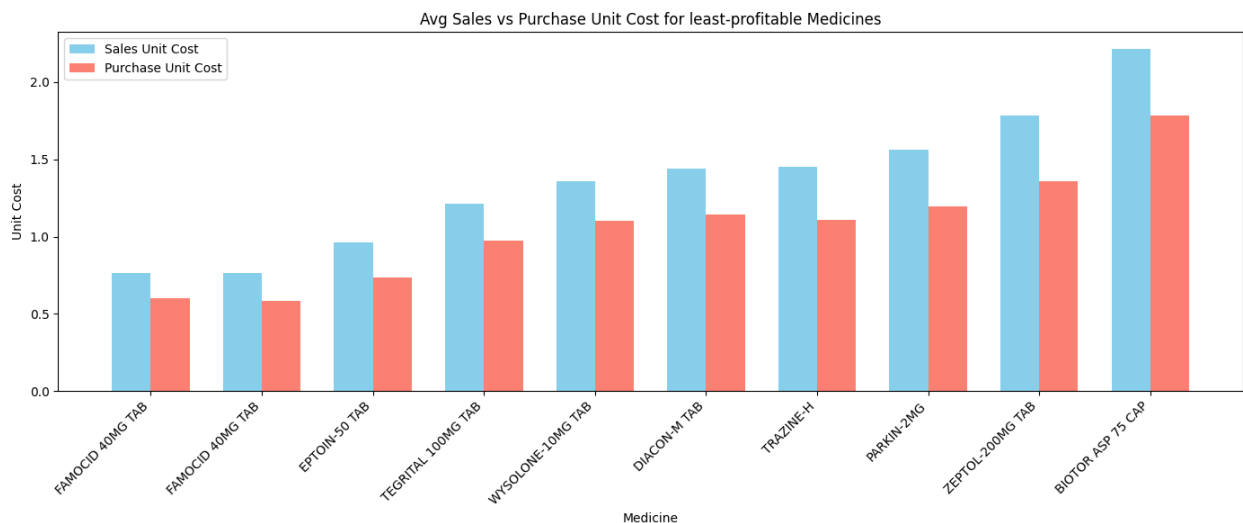


Figure 6: Sales and purchase unit costs for least profitable medicines

Figure 6 represents comparison of the average unit sales cost (blue bars) versus the average unit purchase cost (red bars) for some of the least profitable medicines. Although all products show a higher sales price than purchase cost, the difference (profit margin) is minimal. FAMOCID 40MG TAB and EPTOIN-50 TAB have a very slim margin (less than ₹0.2). Even relatively higher-priced medicines like BIOTOR ASP 75 CAP have limited margin when scaled. This aligns with the bottom 20 profitability chart, showing that despite being sold above cost, low unit price and slim margins contribute to very low overall profit. These medicines may not be viable for retail profitability unless sold in large volumes.

Profitability based on total sold quantity and profit of product

High unit profit does not always mean high overall profit. Some low-margin items might bring higher total profit due to bulk sales. Top profit drivers should be prioritized for inventory and promotional strategies.

Figure 7 and 8 represents the lollipop horizontal chart with x axis as total profit (quantity sold * profit per unit) gained by the pharmacy in the given data duration (6 months) and y axis as the medicines names.

The 20 most profitable chart Figure 7 shows that GLISIMET M2 (14) | BU23297A and GLI-M TAB | SPA231561 implying repeat customers or high sales volume. Profitability here seems driven by both unit profit and quantity sold.

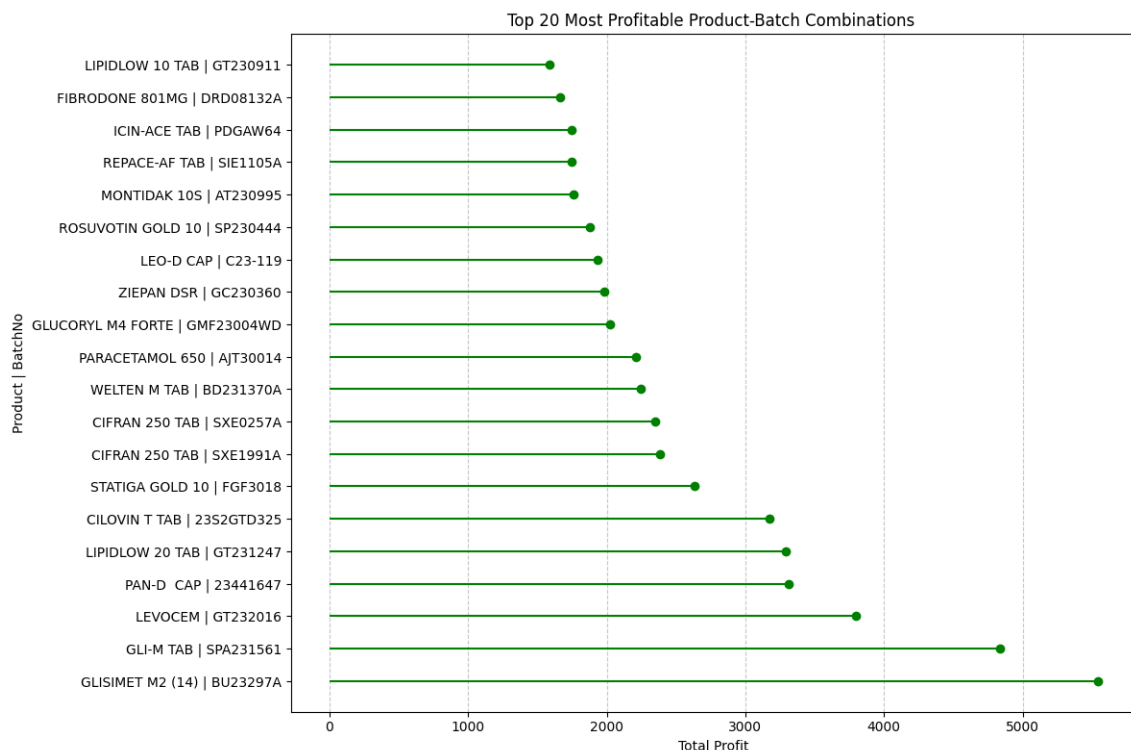


Figure 7: 20 most profitable products based on profit gained

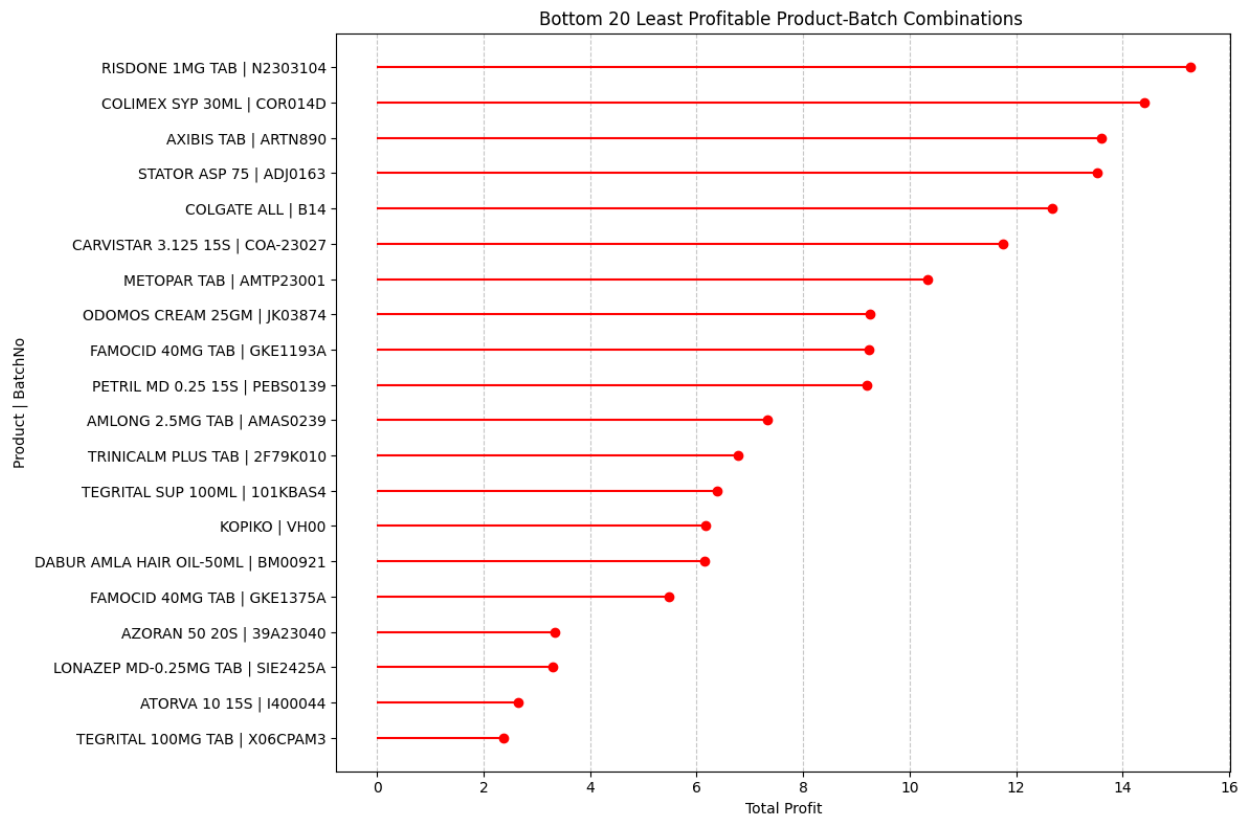


Figure 8: 20 least profitable products based on profit gained

The 20 least profitable product chart Figure 8 shows that these products have extremely low total profits, e.g., TEGRITAL 100MG | X06CPAM3 with under ₹3 in profit. Many of these are slow-moving or low-margin products.

By comparing the gained total profit charts and profit per unit cost charts, some products appeared in the profit per unit chart but may not appear in the total profit chart — indicating low quantity sold. Together, they reveal that while some products are best suited for maximizing profit per transaction, others excel in driving profitability through volume—underscoring the need for a dual approach that balances high-margin items with high-demand offerings.

Stock Status and Non-Availability Forecast

While merged sales data and purchase data quantities to compute current stock levels by Product and BatchNo, With help from the inventory data, validated current inventory against the actual stock in hand.

Products with high sales but low current stock were flagged for stockout risk.

The ROP gives threshold stock level at which you should reorder to avoid running out. It usually matches the average daily usage if calculated for just-in-time restocking. The calculated ROP for all selected data is,

	Product	avg_daily_usage	ROP	current_stock	Reorder_Needed
118	EPTOIN-50 TAB	150.000000	150.000000	300.0	False
59	CIFRAN 250 TAB	106.044776	106.044776	580.0	False
218	NEUROBION FORTE 30,S	90.368421	90.368421	616.0	False
93	DAONIL 5 MG TAB 30,S	81.198020	81.198020	689.0	False
359	THYRONORM-25 TAB PCS	71.179487	71.179487	436.0	False
...	...	...	...	...	...
320	SKINLITE CREAM 15GM	1.000000	1.000000	2.0	False
43	BRIO EYEDROP	1.000000	1.000000	2.0	False
128	FLOJO EYE DROPS	1.000000	1.000000	2.0	False
129	FLOJO-KT DROP	1.000000	1.000000	1.0	False
131	FLUTROL 100 ROTOCAPS	1.000000	1.000000	3.0	False

398 rows x 5 columns

Table 10: Re-order Point

In Table 10, ROP is for Re-order point and the current stock is from the inventory and sales and purchase quantities. Column Reorder_needed provides

True => $\text{current_stock} < \text{ROP}$

False => $\text{current_stock} > \text{ROP}$

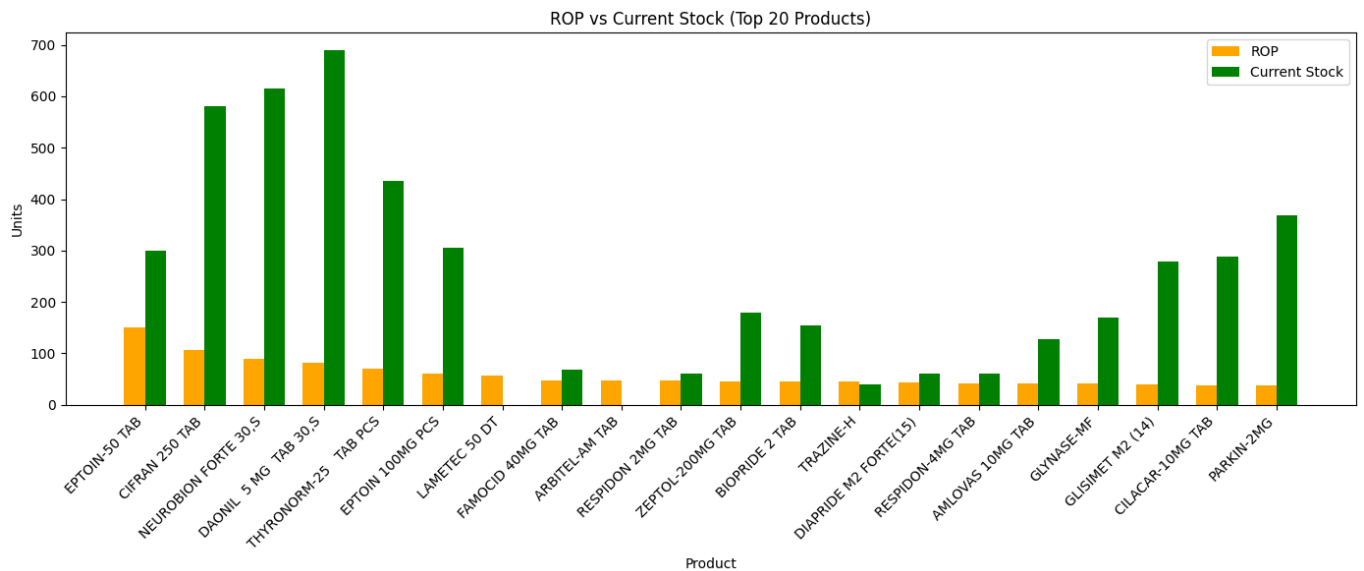


Figure 7: ROP and current stock of top 20 products

The bar chart figure 9 compares the Reorder Point (ROP) and Current Stock for the top 20 products, offering insights into inventory status. For all products shown, the current stock (green bars) exceeds the ROP (orange bars), indicating that none of these items are in immediate need of restocking. Products like DAONIL 5 MG TAB, NEUROBION FORTE, CIFRAN 250 and THYRONORM-25 TAB show particularly high stock levels relative to their reorder points, suggesting potential overstocking that might tie up storage space or capital. On the other hand, products such as FAMOCID 40MG, RESPIDON-2MG TAB, and DIAPRIDE M2 FORTE have relatively smaller margins above their ROP, and while not currently critical, they should be monitored closely to prevent

future shortages. TRAZINE-H is the only product in these 20 medicines which needs an immediate restocking. Overall, inventory management for these items appears proactive, but further optimization may help reduce excess stock for low-risk products.

Economic Order Quantity (EOQ)

Using EOQ formula estimated optimal order sizes for each product to minimize holding and ordering costs. A bar chart compared EOQ across products.

	Product	annual_demand	UnitCost	holding_cost	EOQ
0	ACENOMAC 2 MG	90.0	9.560143	0.956014	137.215781
1	ACIVIR 400DT	73.0	11.810917	1.181092	111.182037
2	ADULT WIPES 10S	6.0	379.995833	37.999583	5.619546
3	AGPENTIN-100MG	60.0	7.268333	0.726833	128.491147
4	ALPRAX 0.25MG TAB	111.5	1.792000	0.179200	352.763326

Table 11: EOQ

This EOQ calculation (Table 11) for all products will helps determine the ideal order quantity that minimizes both ordering and holding costs, thus optimizing inventory management.

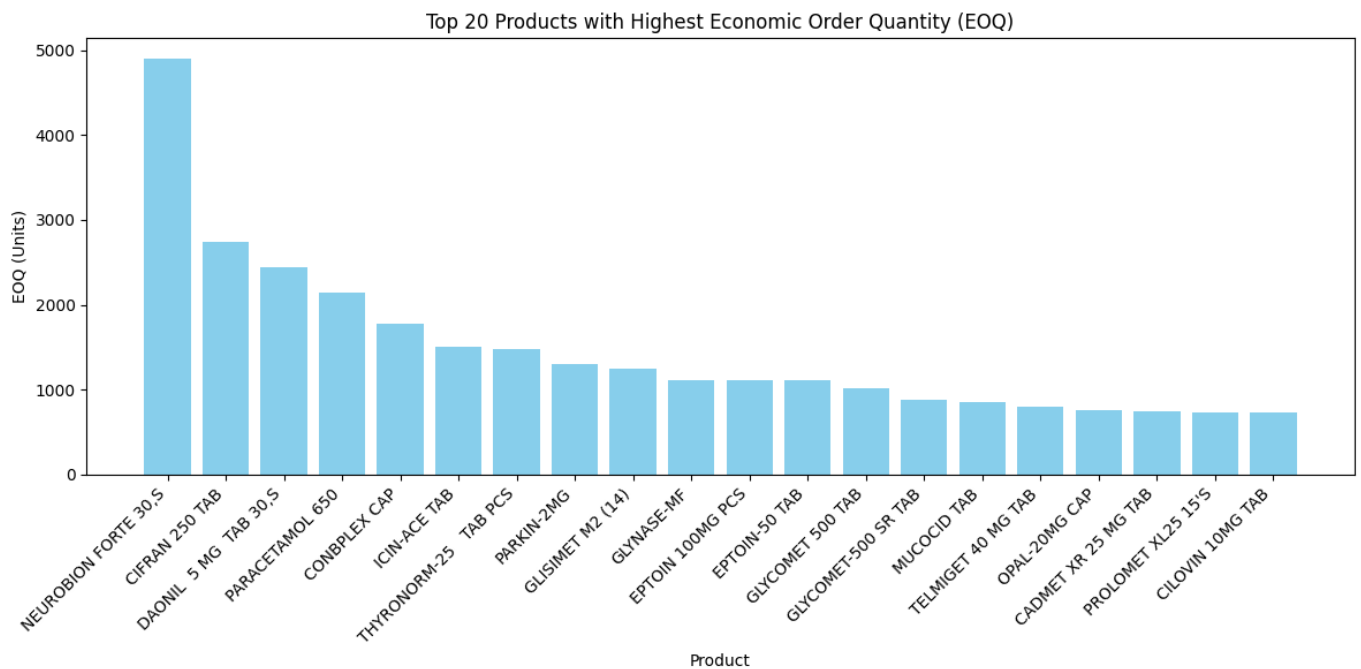


Figure 10: EOQ of top 20 products

The bar chart in figure 10 displays the top 20 pharmaceutical products with the highest Economic Order Quantity (EOQ), which represents the optimal number of units to order each time to minimize total inventory costs. NEUROBION FORTE 30,S stands out significantly with an EOQ of nearly 5000 units, indicating it likely has a combination of high annual demand and low holding cost, making bulk ordering most efficient. It is followed by CIFRAN 250 TAB and DAONIL 5 MG TAB 30,S, both with EOQs exceeding 2500 units, reinforcing their high turnover and cost-effectiveness in larger

procurement sizes. The EOQs gradually decline across the rest of the products, with items like CILOVIN 10MG TAB and PROLOMET XL 25 15'S falling below 800 units. This variation highlights the importance of customized inventory strategies: while some products benefit from bulk purchases, others are better managed with smaller, more frequent orders due to either lower demand or higher holding costs.

Sales Forecasting

Forecasting future demand is critical to avoid stockouts or overstocking. Applied three forecasting techniques:

Moving Average: Smoothed past 3 months data over a window. It forecasted quantity for next 3 months (April, May, June) for each products shown in Table 12.

	Product	ForecastMonth	ForecastQty
0	ACENOMAC 2 MG	2024-04	33
1	ACENOMAC 2 MG	2024-05	33
2	ACENOMAC 2 MG	2024-06	33
3	ACIVIR 400DT	2024-04	35
4	ACIVIR 400DT	2024-05	35

Table 8: Forecasting based on Moving Average

ARIMA: Captured trends, seasonality, and autoregressive components. By using the ARIMA method, forecasted the sales quantity for April, May, June.

	Product	ForecastMonth	ForecastQty
0	ACENOMAC 2 MG	2024-04	30
1	ACENOMAC 2 MG	2024-05	30
2	ACENOMAC 2 MG	2024-06	30
3	ACIVIR 400DT	2024-04	78
4	ACIVIR 400DT	2024-05	77

Table 13: Forecasting based on ARIMA

Exponential Smoothing: Weighted recent data more heavily, good for data with consistent trends. Using exponential smoothing forecasted the products sales quantity for April, May, June same as the above 2 forecasting methods.

	Product	ForecastMonth	ForecastQty
0	ACENOMAC 2 MG	2024-04	32
1	ACENOMAC 2 MG	2024-05	33
2	ACENOMAC 2 MG	2024-06	33
3	AGPENTIN-100MG	2024-04	30
4	AGPENTIN-100MG	2024-05	35

Table 14: Forecasting based on Exponential Smoothing

The forecast comparison for ACENOMAC 2 MG across three different models—Moving Average

(Table 12), ARIMA (Table 13), and Exponential Smoothing (Table 14)—reveals distinct patterns. The Moving Average model provides a consistent forecast of 33 units for April to June 2024, indicating a stable outlook without sensitivity to recent changes. In contrast, the ARIMA model forecasts a steady but lower value of 30 units, suggesting a more conservative prediction, possibly based on past declining trends or noise filtering. The Exponential Smoothing model shows a slight upward adjustment, starting at 32 units in April and rising to 33 units in May and June, reflecting its ability to adapt to recent data trends. While Moving Average offers stability and ARIMA shows caution, Exponential Smoothing provides a balanced forecast that reacts modestly to changes, making it potentially more accurate in a gradually rising demand scenario.

Considering a product named ACENOMAC 2 MG to visualise the difference in forecasting of a medicine for the next 3 months (April, May, June) based on 3 forecasting methods Moving average, ARIMA, Exponential smoothening.

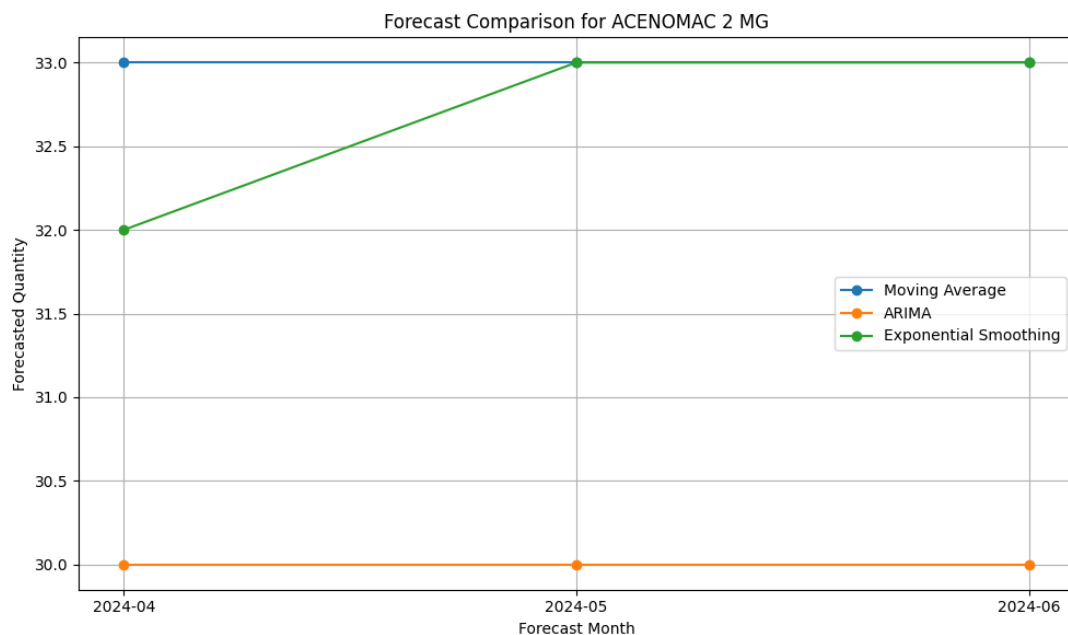


Figure 11: Comparison of 3 forecasting based on 3 methods for the product ACENOMAC 2 MG

The line chart (figure 11) illustrates a forecast comparison for the product ACENOMAC 2 MG across three months—April, May, and June 2024—using three different forecasting models: Moving Average, ARIMA, and Exponential Smoothing. The Moving Average model predicts a constant quantity of 33 units for all three months. Similarly, the ARIMA model forecasts a steady 30 units throughout the period. In contrast, the Exponential Smoothing method shows a slight upward trend, starting at 32 units in April and increasing to 33 units from May onward. This indicates that while both Moving Average and ARIMA models assume stability, Exponential Smoothing incorporates recent trends, reflecting a modest growth. Overall, Moving Average and Exponential Smoothing converge to the same forecast in May and June, while ARIMA remains consistently lower, potentially underestimating demand.

4. Interpretation of Results and Recommendation

Interpretation of Results:

Profitability Trends:

The profit per unit chart highlights certain medicines with extremely high margins. However, high per-unit profit doesn't always translate to overall profitability if sales volume is low. The total profit chart provides a clearer picture of real business impact. It shows which product-batch combinations truly contributed to the business's revenue—balancing both margin and quantity sold.

Inventory and Stock Levels:

Current stock vs ROP analysis shows a few critical medicines are below their reorder points, signaling a risk of stockouts. EOQ analysis revealed which medicines should be ordered in bulk to reduce ordering costs while keeping stock at an optimal level.

Sales Forecasting:

Forecasting using Moving Average, ARIMA, and Exponential Smoothing gave consistent trends for high-demand products. For example, ACENOMAC 2 MG showed similar patterns across models, confirming its stable demand. These methods provided future demand estimates, crucial for pre-planning purchases and avoiding both stockouts and overstocking.

Recommendations:

1) Focus on High-Profit + High-Volume Products:

Prioritize procurement and stock availability for products that have both good profit per unit and high overall profit. Monitor products with high margin but low sales to explore if promotion, discounting, or better placement could increase volume.

2) Improve Stock Planning Using EOQ and ROP:

Use EOQ values to decide batch sizes for restocking especially for fast-moving and consistent-demand products. Use ROP flags to trigger purchase orders before stockouts occur. Automate this process if possible.

3) Discontinue or Reassess Low-Profit Products:

Products frequently appearing in the bottom 20 least profitable (both in unit and total profit) should be evaluated for Supplier cost renegotiation, Price adjustments and Replacement with better-performing alternatives

4) Leverage Forecasting for Smarter Purchasing:

Use the forecasted demand to create a rolling 3-6 month stock plan. For highly predictable products, ARIMA and Exponential Smoothing models are reliable and should be integrated into regular planning.

5) Manage Brand Diversity Effectively:

Reduce brand clutter by focusing on top-performing brands of the same generic. This lowers inventory cost and improves shelf space efficiency. Eliminate or limit brands that consistently underperform in terms of both sales volume and profit.

Maintain regular brand-level performance reports to identify best-sellers vs. stagnant products. Use this insight to shape future purchase decisions and avoid tying up capital in slow-moving brands.

6) Capture Patient Brand Preferences:

Keep a simple excel record or register to log cases where patients ask for a specific brand that is not currently stocked. This helps capture real-world demand influenced by doctor prescriptions especially important because many patients trust only the prescribed brand, even if generics exist. Frequent non-availability of popular brands can lead to lost sales and customer dissatisfaction, so this insight should influence restocking.

7) Regular Data-Driven Review:

Repeat this profitability and stock analysis monthly or quarterly to catch trends early. Create dashboards for real time monitoring of stock, profit, and forecasted demand to support faster, more informed decisions.